

AAYUSH KALIA

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EDUCATION

Guru Nanak Dev Engineering College, Ludhiana
B.Tech in Computer Science & Engineering

June 2026
CGPA: 7.70

EXPERIENCE

Centralize 360

Jan 2026 – Present

Full Stack Intern

- Built an Electron.js desktop app to automate Tally data extraction and backend sync, shipped as a client deliverable
- Developed REST APIs (Node.js, Express) and React frontend integration for end-to-end data flow, handling mapping, validation, and transformation
- Diagnosed and resolved cross-layer failures including silent DB persistence bugs, Sequelize ORM errors, and schema mismatches across Electron, API, and database
- Evaluated batch vs real-time ingestion trade-offs for financial data pipelines, informing architecture decisions on performance and reliability

GyanDhan

June 2024 – August 2024

Marketing and Outreach Intern

- Conducted market research and executed outreach campaigns, increasing webinar attendance by 35%

TECHNICAL SKILLS

Web/Backend: JavaScript, TypeScript, Node.js, Express.js, React, Electron.js, RTK Query, MySQL, PostgreSQL, Sequelize, JSON, Git, GitHub, VS Code

ML/Data: Python, C, C++, TensorFlow, Keras, PyTorch, Scikit-learn, SHAP, Pandas, NumPy

PROJECTS

Skin Disease Classification using Explainable AI

Major Project

- Built a 6-class skin disease classification system using EfficientNetB0 with transfer learning, achieving 89% test accuracy and 96% validation accuracy.
- Applied Explainable AI techniques (Grad-CAM, SHAP, LIME) and built a full TensorFlow/Keras pipeline with Dropout and Softmax for multi-class generalization.
- Awarded 1st Prize, CSE Department for innovation and real-world healthcare impact.

PTU Sports Meet Registration Portal

Frontend Developer

- Designed and developed a responsive React.js web application managing 350+ student registrations.
- Integrated dynamic jersey allocation and auto-generated event records, reducing manual workload by 70%.

Multimodal & Multilingual Hate Speech Classifier

Minor Project

- Engineered a hybrid model using BERT for text and ResNet for image inputs to detect hate speech.
- Processed a multilingual dataset (English, Hindi, Hinglish) of 5,000 samples, achieving an F1-score of 87% using early fusion of visual and textual modalities.

VOLUNTEERING AND LEADERSHIP

Training and Placement Cell

March 2024 – Present

Student Moderator

- Contacted 100+ HR professionals for campus drives, coordinated interview logistics, and presented placement analytics reports to faculty.